

Solving a linear system

Question 1 / 10 points

Solve $2x + y = 5$ and $x - y = 1$ using row reduction.

Show the intermediate operations.

Starting matrix

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array} \right]$$

Swap R1 and R2

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 2 & 1 & 5 \end{array} \right]$$

$R2 \leftarrow R2 + (-2) R1$

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 3 & 3 \end{array} \right]$$

$R2 \leftarrow (1/3) R2$

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & 1 \end{array} \right]$$

$R1 \leftarrow R1 + (1) R2$

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right]$$

Final answer: $x = 2, y = 1$

Equivalent valid row-reduction sequences are accepted.